The students that wrote down intermediate stages in their calculations could gain a mark for their working eg √13.82 = 3.717....

Some students lost the mark for part (b) by writing 1.8, rather than 1.84. It was pleasing to see that many students gained the mark in part (b) by correctly rounding their answer, even if incorrect, from part (a).

Q3.

This question was not well answered. Many students gained some reward for dividing the volume (750) by the length (25) but were then unable to equate their result to the area of the cross-section. It was rare to find any formula given for the area of the cross-section or volume. An answer of 6 was awarded one mark for those students incorrectly working with a cuboid.

Q4.

This question was not well answered. Some students gained a mark, usually for substituting 3 into the given formula for the surface area of the sphere. Some gave a numerical value for pi and attempted to find the surface area of the sphere, with varied success.

Q5.

Neither part of this question was well answered and there were a disappointing number of blank responses.

A few students indicated the recognition of the need to use trigonometric ratios. Of these, some wrote down SOHCAHTOA or drew "formula triangles" but were not able to select the most appropriate ratio to use.

Students who correctly identified they needed to use the tangent ratio in part (a) normally scored both marks. When only 1 mark was awarded, the final mark was often lost due to poor application of the calculator for example the calculator was not in degree mode.
The most common incorrect approach seen was to use the sum of the angles in a triangle by subtracting 90°and 34° from 180°, despite the question requiring a missing length to be calculated. Some attempted to use Pythagoras, squaring the 34 and the 12, mixing angles and side lengths was seen.

Part (b) was very similar to part (a) in the quality of response. For those who realised the need to use trigonometry many did not identify the correct ratio to use or know that the inverse function

was needed in order to find an angle. Although $\frac{15}{18}$ = 0.83 was sometimes calculated and formed part of the working, this was insufficient for credit to be given as a correct equation involving cos *x* was required.

Some students gave an answer of 34 which had evidently come from assuming the angles in the diagram from part (a) were the same as 90 - 56 was also seen as the working.

Less often seen was an answer of 33 from adding the side lengths. Another common incorrect answer of 45 came from assuming the triangle is isosceles. Some attempted to use Pythagoras's Theorem but were unable to complete the alternative approach far enough to be credit worthy.

Q6.

Many students scored both marks on this question correctly drawing the reflection in the mirror line. Only a few students drew a correct reflection using an incorrect mirror line to score the method mark only.

It was disappointing to see responses that scored no marks, the most common error was to attempt a rotation of the shape or a translation of the original shape to the bottom left-hand side of the grid. These incorrect transformations scored no marks.

Q7.

This question was well answered, with most students scoring the 2 marks available. The most common incorrect responses were either to draw a 6cm by 3cm rectangle, seemingly from counting the grid lines rather than the squares, or to draw either a 5cm by 1cm rectangle or a 5cm by 5cm square. When full marks were not awarded, a mark was often achieved for engaging with the scale given by either drawing a diagram with one dimension correct or by calculating both dimensions correctly.

Q8.

In part (a) most students used a non-calculator method and obtained the correct answer. It was obvious when calculators were used where an answer of 196 ÷ 603 = 3 h 26 min was seen. Many gained 1 mark for getting as far as 3 hours.
In part (b) it was very rare to see any correct answers. Some wrote down the formula triangle for speed distance but could not relate this to an algebraic expression. The closest many students got was 2*x*, the most common incorrect algebraic answer.

Q9.

In this cohort, very few students showed any understanding of $y = mx + c$
Closest attempts were 3 or 4 in part (a) and (3, 4) or (3, -4) in part (b).

Q10.

More students were successful on this question. The vast majority were able to correctly order the decimal figures given.

Q11.

Ordering of negative numbers was correctly carried out by most students.

Q12.

The vast majority of students answered both parts of this question correctly, demonstrating a good understanding of the skills being assessed.

In part (a) many did gain the first mark by stating an answer within the range of 14 to 16. A common error seen was to read off the value of 75 for 5 hours but not make the adjustment to a payment for one hour or not being able to interpret the divisions on the scale correctly as £2 per square, with £10.60 being seen frequently. A very small number of students added together readings for 1, 2, 3, 4 and 5 hours.

The most common and successful approach to part (b) was to use their hourly rate found in part (a) and multiply by 36. A build up method using one or more readings from the graph was less common and less successful, particularly as many using this method only found the pay for 35 hours by using the reading at 5 hours and not adding an extra hour of pay to show a complete build-up.

Q13.

The first two parts of this question were not answered well. $2ad$ or $2a + d$ were common incorrect answers in part (a), whilst in part (b) very few demonstrated any understanding of factorisation. $y$ and $y^2$ were often incorrect answers offered here.

Students had greater success with part (c), gaining full marks for an answer of 11. Many students showed the value of $x$ as 11 in their working but then gave an answer of 44 on the answer line. Very few students used algebra explicitly in the question and many took the wrong first step by subtracting instead of adding 7.

Q14.

In part (a) many students correctly expanded the first part of the bracket to get $y^2$ but did not then multiply $y$ by 5 to obtain $5y$.

Part (b) was less well answered. An incorrect factor of 4 was commonly seen outside the bracket.

In part (c) more students found success. It was pleasing to see the layout of the process to solve the equation clearly structured. Some students were more confident at expanding brackets than at solving equation algebraically, though some who expanded incorrectly were able to access the mark for rearranging their equation ready for solution. Unsuccessful efforts included those who tried a trial and improvement approach of substituting a variety of numbers into the equation.

Responses to part (d) were varied. Some secured full marks, but rarely. Most commonly students processed the numerical term but then multiplied the indices. Some failed to recognise that the lone $e$ or $f$ had an index of 1. Those who correctly remembered the rules of indices frequently wrote $9e^3f^4$

Q15.

In part (a) many argued that Jasper was not correct because $x$ could not be a decimal. The few who did substitute $x = 0.5$ into the given expression usually got the right idea and correctly explained the error. Only a very small minority gained any credit in part (b). Use of algebra was very nearly non-existent with students instead preferring to attempt numerical methods which were usually doomed to failure. The most common incorrect approach was to assume all 4 sides of the kite were equal and divide 64 by 4. A few students gained a mark for writing $3(4x – 2)$ or $12x – 6$, some then went on to incorrectly equate that to 64. Some thought they were dealing with angles and introduced "360 degrees" into their working. Even when students correctly calculated $AD$, they were then unable to link this to an equation or expression for perimeter. Lots of incorrect responses featured students simply setting $4x – 2 = 64$ and then attempting to solve

from there; this gained no credit.

Q16.

Few failed to get the correct answer of 14 when dividing 42 by 3 although a small minority insisted on halving then halving again thinking this was the correct method of finding a third of a quantity. Many students also used repeated addition or listed multiples, the majority eventually arriving at 42.

## Mark Scheme

Q1.

| Question | Answer | Mark | Mark scheme | Additional guidance |
|---|---|---|---|---|
| (i) | 21 | M1 | for 180 – 75 – 84 | Angle may be indicated on the diagram |
| | | A1 | cao | |
| (ii) | Reason given | C1 | for reason that Angles on a straight line add up to 180 | The key words underlined must be present<br>There should be no incorrect reasons given |

Q2.

| Question | Answer | Mark | Mark scheme | Additional guidance |
|---|---|---|---|---|
| (a) | 1.844977205 | M1 | for 3.403(940887)<br>or 3.717(526059)<br>or 2.014(944168)<br>or 1.84(...) or<br>1.8(...) | Accept consistent use of a comma to indicate a decimal point |
| | | A1 | for 1.844(977205) | Answer must be given to at least 3 decimal places rounded or truncated |
| (b) | 1.84 | B1 | for 1.84 or ft from (a) provided answer to (a) has at least 3 dp | |

Q3.

| Question | Answer | Mark | Mark scheme | Additional guidance |
|---|---|---|---|---|
| | 12 | P1 | for a process to find the area of cross section,<br>eg $750 \div 25$ (= 30) oe<br>**or** $\frac{1}{2} \times 5 \times h$ oe | May use any letter for $h$ or may use ? |
| | | P1 | for a correct equation in $h$,<br>eg $750 \div 25 = \frac{1}{2} \times 5 \times h$ oe<br>**or** $\frac{1}{2} \times 5 \times h \times 25 = 750$ oe<br><br>**or** for a complete process to find $h$,<br>eg. $\frac{750}{25} \times \frac{2}{5}$ oe **or** "30" $\times 2 \div 5$ | |
| | | A1 | cao<br><br>SC B1 for answer of 6 if P0 scored | |

Q4.

| Question | Answer | Mark | Mark scheme | Additional guidance |
|---|---|---|---|---|
| | Shown | M1 | for a correct expression for the area of one face of the cube, eg. $x^2$<br>**or** a correct expression for the surface area of the cube, eg $6 \times x^2$ | No marks for $x = \sqrt{6\pi}$ without any working. |
| | | M1 | for a correct expression for the surface area of the sphere, eg $4 \times \pi \times 3^2$ ($= 36\pi$) | |
| | | M1 | for forming a suitable equation, eg $6 \times x^2 = 4 \times \pi \times 3^2$ **or** $6x^2 =$ "$36\pi$" | $6 \times x^2 = 4 \times \pi \times 3^2$<br>$x^2 = 36\pi \div 6$<br>$x = \sqrt{6\pi}$ |
| | | A1 | for completing the method to $x = \sqrt{6\pi}$ or $k = 6$ | |

Q5.

| Question | Answer | Mark | Mark scheme | Additional guidance |
|---|---|---|---|---|
| (a) | 17.8 | M1 | for $\tan 56 = \frac{x}{12}$ or<br>$(BC) = 12 \times \tan 56$ oe<br>**or** alternative method to find $BC$ | For any alternative method candidates must arrive at an equation with BC as the only unknown |
| | | A1 | for an answer in the range 17.7 to 17.8 | If an answer in the range 17.7 to 17.8 is given in the working space then incorrectly rounded, award full marks. |
| (b) | 33.6 | M1 | for $\cos x = \frac{15}{18}$ **or** $\cos x = 0.83..$<br>**or** $x = \cos^{-1}\frac{15}{18}$<br>**or** alternative method to find $x$ | For any alternative method candidates must arrive at an equation with $x$ as the only unknown |
| | | A1 | for an answer in the range 33.5 to 33.91 | If an answer in the range 33.5 to 33.91 is given in the working space then incorrectly rounded, award full marks. |

Q6.

| Question | Answer | Mark | Mark scheme | Additional guidance |
|---|---|---|---|---|
| | Reflection | M1<br><br>A1 | for a correct reflection of the shape in any horizontal line other than the given mirror line<br>for a fully correct reflection | Allow free hand drawing |

Q7.

| Question | Answer | Mark | Mark scheme | Additional guidance |
|---|---|---|---|---|
| | 7cm by 4cm rectangle drawn | M1 | for a rectangle drawn with one correct dimension<br>**or** 35 ÷ 5 (=7)<br>and 20 ÷ 5 (=4) | Correct calculations/measurements seen the method mark can be awarded even if the drawing is incorrect or not present<br>Accept any orientation of a correct rectangle |
| | | A1 | for a fully correct 7cm by 4cm rectangle drawn | |

Q8.

| Question | Answer | Mark | Mark scheme | Additional guidance |
|---|---|---|---|---|
| (a) | 3 hrs 16 mins | P1 | 196 – 60 – 60 – 60 (=16) oe **or**<br>196 ÷ 60 (= 3.26.. or 3.27…)<br><br>**or** states 3 hours in their answer (with an incorrect number of minutes or minutes left blank) | |
| | | A1 | 3 hours 16 minutes | |
| (b) | $\frac{x}{2}$ | B1 | $\frac{x}{2}$ oe | |

Q9.

| Question | Answer | Mark | Mark scheme | Additional guidance |
|---|---|---|---|---|
| (i) | −4 | B1 | cao | |
| (ii) | (0, 3) | B1 | cao | |

Q10.

| Question | Answer | Mark | Mark scheme | Additional guidance |
|---|---|---|---|---|
| | 0.12, 0.21, 1.02, 1.20 | B1 | accept 1.20, 1.02, 0.21, 0.12 | |

Q11.

| Question | Answer | Mark | Mark scheme | Additional guidance |
|---|---|---|---|---|
| | –10, –7, –2, 0, 1, 8 | B1 | Accept the reverse order, eg 8, 1, 0, –2, –7, –10 | |

Q12.

| Question | Answer | Mark | Mark scheme | Additional guidance |
|---|---|---|---|---|
| (a) | 15 | B1 | 14 to 16 | |
| (b) | 540 | M1 | for a complete method, eg 30 × (36 ÷ 2) **or** 45 × (36 ÷ 3) **or** 60 × (36 ÷ 4) **or** ft "hourly rate from (a)" × 36 | May be seen using a complete build up method<br>for "45" allow 44 to 46 ft for accuracy |
| | | A1 | for 540 **or** ft (a) | Condone use of mixed rates eg 75×7 + 16 =541 |

Q13.

| Question | Answer | Mark | Mark scheme | Additional guidance |
|---|---|---|---|---|
| (a) | $2a + 2d$ | B1 | cao | Accept $2 \times a + 2 \times d$ |
| (b) | $y(6y - 5)$ | B1 | cao | Accept $y \times (6y - 5)$ |
| (c) | 11 | M1 | for isolating $x$ terms, eg $4x = 37 + 7$ or $4x = 44$<br>**or** for $x - \frac{7}{4} = \frac{37}{4}$<br>**or** for $37 + 7 = 44$ followed by "44" ÷ 4 (= 11) | |
| | | A1 | cao | |

Q14.

| Question | Answer | Mark | Mark scheme | Additional guidance |
|---|---|---|---|---|
| (a) | $y^2 + 5y$ | B1 | cao | |
| (b) | $2(2a - 3)$ | B1 | cao | |
| (c) | 2.9 | M1 | for a correct first stage<br>eg. expanding the brackets,<br>$2 \times 5x - 2 \times 4$ (= $10x - 8$)<br>**or** division of both sides by 2, eg<br>$\frac{2(5x-4)}{2} = \frac{21}{2}$ | |
| | | M1 | for isolating terms in $x$ eg $10x = 21 + 8$ | |
| | | A1 | oe | |
| (d) | $20\,e^3 f^4$ | M1 | for any two of $4 \times 5$ (=20), $e^{2+1}$ (=$e^3$), $f^{1+3}$ (=$f^4$) in a product or written as individual terms | Do not award if there is contradiction |
| | | A1 | cao | |

Q15.

| Question | Answer | Mark | Mark scheme | Additional guidance |
|---|---|---|---|---|
| (a) | Explanation | C1 | for explanation, eg $AB$ cannot be zero (cm) **or** shows $AB$ to be zero, eg $4 \times 0.5 - 2 = 0$ | Accept say '$AB$ would then be 0' |
| (b) | 2.5 | P1 | for a correct expression for $AD$, eg $3(4x - 2)$ **or** $12x - 6$<br><br>**OR** $2(3AB + AB) = 64$ oe<br>**or** $3AB + AB = 32$ oe **or** $AB = 8$<br><br>**OR** for an equation with mixed variables, eg. $6AB + 2(4x - 2) = 64$ | May be seen on diagram |
| | | P1 | for forming a correct equation in $x$, eg $4x - 2 + 4x - 2 + 3(4x - 2) + 3(4x - 2) = 64$<br>**or** $4x - 2 = 8$<br>**or** $4x - 2 + 3(4x - 2) = 32$ | |
| | | A1 | cao | |

Q16.

| Question | Answer | Mark | Mark scheme | Additional guidance |
|---|---|---|---|---|
| | 14 | M1 | for $42 \div 3$ | |
| | | A1 | cao | |